

Vishal Saini

Software Engineer | Backend & AI Systems

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Professional Summary

Software Engineer with experience building scalable healthcare, AI-powered and multi-tenant applications using Node.js, Express.js, MongoDB, Redis and RabbitMQ. Developed EMR systems, AI-assisted diagnostic pipelines, semantic search platforms and speech-to-text analytics workflows using Gemini AI, OpenAI Whisper and Qdrant. Skilled in REST APIs, workflow automation and scalable backend architecture.

Technical Skills

Core Languages & Frameworks: Node.js, Express.js, JavaScript (ES6+), React.js, Python (Flask), Java.

AI & Natural Language Processing (NLP): Generative AI, OpenAI APIs (GPT, Whisper), Gemini AI APIs, Prompt Engineering, AI Workflow Automation, Speech-to-Text Processing.

Architecture & Backend Infrastructure: Multi-tenant Systems, RESTful APIs, System Design, Role-Based Access Control (RBAC), Microservices, Message Queues (RabbitMQ), Distributed Caching (Redis).

Databases & Vector Search: MongoDB, MySQL, Qdrant (Vector Database), Semantic Search, Data Modeling.

DevOps & Tools: Docker, CI/CD Pipelines, Linux, Git, GitHub, Postman, AWS, GCP.

Professional Experience

ICA (Impetus Consulting Associates) Pvt. Ltd.
2025 – Present

Jaipur, Rajasthan | August

- Working on scalable healthcare and AI-driven backend applications using Node.js, React.js, MongoDB, Redis, and RabbitMQ.
- Developing a multi-tenant Patient Management System (PMS) with secure patient data handling, appointment workflows, and role-based access systems.
- Built REST APIs and backend workflows for patient history management, online prescriptions, and healthcare dashboard integrations.
- Integrated Gemini AI pipelines to process medical reports and assist doctors with faster patient data understanding and automated summaries.
- Designed optimized MongoDB schemas and scalable backend architecture for handling high-volume healthcare records and real-time operations.
- Contributed to workflow automation and backend optimizations to reduce manual operations and improve system efficiency.

PhysioPlus Healthcare Pvt. Ltd.
August 2025

Jaipur, Rajasthan | March 2025 –

- Developed scalable backend services and REST APIs for healthcare operational workflows and appointment systems.
- Designed and implemented a location-based physiotherapist recommendation engine using latitude/longitude coordinates and the Haversine formula.

- Improved notification workflows and real-time appointment updates for better patient engagement and operational efficiency.
- Worked on backend optimization, deployment workflows, and Dockerized environments for smoother CI/CD operations.

Key Projects

Auto Call Auditor (ACA) System

Tech Stack: Node.js, Express.js, OpenAI Whisper API, Gemini AI, RabbitMQ, React.js

- Built an AI-powered call analysis platform to process customer-agent conversation recordings and generate automated performance insights.
- Developed an asynchronous audio processing pipeline using OpenAI Whisper API to convert call recordings into structured text transcripts.
- Integrated Gemini AI to analyze customer conversations, evaluate agent communication quality, identify customer sentiment, and generate detailed KPI-based performance reports.
- Implemented AI workflows to measure metrics such as communication behavior, customer handling quality, conversation effectiveness, and response patterns.
- Used RabbitMQ for background processing of large audio uploads, ensuring smooth and scalable asynchronous task handling without blocking backend services.
- Built dashboards for displaying call analytics, AI-generated summaries, agent scores, and conversation insights.

Patient Management System (PMS)

Tech Stack: Node.js, MongoDB, Qdrant, Gemini AI, Redis, React.js

- Developed a scalable Patient Management System for handling appointments, patient records, prescriptions, and healthcare workflows.
- Integrated Gemini AI to analyze patient intake forms and diagnostic reports for automated healthcare insight generation.
- Implemented semantic medical record search using Qdrant vector database for faster internal healthcare data retrieval.
- Integrated Redis caching to optimize repeated AI-generated healthcare summaries and improve response performance.
- Built secure backend APIs and responsive healthcare dashboards for doctors and clinic staff.

Location-Based Recommendation & Admin Infrastructure

Tech Stack: Node.js, Express.js, MongoDB, React.js, RBAC

- Access Security: Authored a secure, centralized administration engine utilizing rigid Role-Based Access Control (RBAC) to govern global CRUD operations across sensitive user databases.
- Spatial Routing: Managed real-time data schemas to coordinate multi-variable matching filters (availability, radius, domain specialization) between patients and specialized clinicians.

Education

MCA - Rajasthan Technical University, Kota
(Expected)

2026 – 2028

BCA – Maharishi Arvind Institute of Science & Management

CGPA: 7.7 / 10